

AI Recitation of the Holy Quran

In Light of Islam

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A Critical Review of Permissibility, Tajweed Errors, and Islamic Jurisprudence

Based on Five Peer-Reviewed Papers and Primary Fatwa Sources

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Abstract

The emergence of artificial intelligence in Islamic practice has introduced a question of serious scholarly weight: when an AI system recites or evaluates Quranic recitation and produces errors, does Islamic jurisprudence treat those errors as permissible or prohibited? This review synthesises five peer-reviewed academic papers, three primary fatwa sources, direct Quranic evidence, and the established historical doctrine of Talaqqi-Musyafahah — the Prophet Muhammad's method of oral Quranic transmission — to construct an evidence-grounded answer.

The majority position, supported by the preponderance of classical principles and contemporary fatwas, concludes that AI recitation errors are not permissible as a standard or substitutive practice. A conditional, minority position permits AI tools exclusively in an assistive capacity, subject to three strict scholarly conditions. This document is structured as a critical review. It progresses from the technical landscape of AI Quran tools, through systematic analysis of the five papers, into Islamic legal reasoning and historical precedent, and culminates in a probability-weighted conclusion.

Note:

Probability estimates herein are a synthesis of the reviewed sources and represent the author's assessment, not empirical data.

1. Introduction

The recitation of the Holy Quran is not merely a linguistic act in the Islamic tradition — it is an act of worship governed by an intricate, centuries-refined system of phonological rules called Tajweed. Every letter in the Quran carries a specific articulation point (Makhraj), a prescribed duration, a set of phonetic interactions with adjacent letters, and an acoustically defined quality (Sifah). The Prophet Muhammad received the Quran from the Angel Jibreel (AS) with these precise characteristics intact, and transmitted them orally to his companions in a process known as Talaqqi (direct reception) and Musyafahah (face-to-face transmission). This chain — unbroken across fourteen centuries — constitutes the methodological backbone of authentic Quranic recitation.

Into this ancient and carefully guarded tradition, artificial intelligence now introduces a new and contested actor. AI-powered applications — ranging from Tarteel AI's real-time mistake detection to TajweedMate's automated pronunciation scoring — claim to evaluate, and in some cases produce, Quranic recitation. These tools are trained on digital audio datasets, acoustic feature extraction algorithms, and deep learning architectures. They carry no chain of transmission, no scholarly license (Ijazah), and no capacity for the contextual, phonologically sensitive correction that a qualified human Qari provides.

This review asks a pointed question: when these AI systems fail — when they misidentify a Makhraj, distort a Mad elongation, or silently omit an Ikhfa' rule — does Islamic law treat that failure as permissible or prohibited? The answer has practical consequences for the estimated 1.8 billion Muslims who use these tools in varying degrees of reliance.

2. Literature Review: Five Foundational Papers

The following five papers form the evidentiary core of this review. Each is analyzed for its direct contributions to understanding AI errors in Quranic recitation and their broader implications.

2.1 Alrumiah & Al-Shargabi (2023) — Springer Nature

Intelligent Quran Recitation Recognition and Verification: Research Trends and Open Issues | Arabian Journal for Science and Engineering (Springer)

This paper constitutes the most rigorous systematic literature review of AI-based Quranic recitation systems published to date, spanning published work from 2006 through 2022. It answers six research questions covering the full pipeline of intelligent recitation systems: the acoustic feature extraction methods used, the machine learning architectures deployed, the error categories targeted, and the open problems that remain unresolved.

The paper's taxonomy of AI techniques is particularly instructive. Traditional approaches — Hidden Markov Models (HMM), Dynamic Time Warping (DTW), and Mel-Frequency Cepstral Coefficients (MFCC) — were the dominant methods through the 2010s. Since 2018, deep learning architectures, particularly Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, have displaced them as the leading paradigms.

Despite this technical progression, the paper identifies several persistent limitations: existing systems degrade significantly under natural recitation conditions; diacritics-based error detection remains an open problem; and sequence recitation errors are largely unaddressed. The gap between 96% accuracy on controlled test sets and real-world recitation performance is not a marginal engineering concern — it is the precise space where theologically consequential errors live.

2.2 Adnan (2024) — Khazanah Journal of Religion and Technology

Leveraging Artificial Intelligence Technologies in the Service of the Holy Quran and Its Sciences | Vol. 2, No. 2

This paper occupies a unique position in the literature because it is the only peer-reviewed study reviewed here that frames the AI-Quran relationship explicitly through the lens of Islamic sciences rather than through computer science or engineering. The author surveys AI applications across six domains of Quranic scholarship: Tafsir (exegetical analysis), Hadith sciences, Quran memorisation, recitation evaluation, fatwa generation, and religious awareness dissemination.

The paper contains a crucially important caveat: AI tools functioning in the domain of Islamic scholarship must be developed and validated through collaboration between technical engineers and qualified Islamic scholars. Without this dual validation, the risk is the production and propagation of Islamically incorrect output under the imprimatur of a digital authority that many users may uncritically accept.

2.3 Fatwa Sources — IslamWeb and IslamQA

IslamWeb Fatwa No. 510187 | IslamQA Fatwa No. 512399 | IslamQA Fatwa No. 391117 (Auto-tune Ruling)

IslamWeb Fatwa No. 510187, specifically addressing the permissibility of AI-based Quran learning tools such as Tarteel AI, delivers the most direct ruling in the literature. The fatwa conditionally permits such tools if qualified Tajweed scholars have certified them as error-free. However, it articulates a principle of particular force: if Islamic tradition holds that the Quran may not be learned from a person who makes recitation mistakes, then it is logically more fitting that it not be learned from a machine that makes mistakes. This constitutes a formal jurisprudential analogy (Qiyas).

IslamQA Fatwa No. 512399 rules that using a human Qari's voice in an AI system requires that Qari's explicit permission, and that the Qari does not receive recitation reward for AI-generated output based on their voice — since the act of recitation requires conscious intention (Niyyah) and physical performance by a living, accountable person.

The fatwa on auto-tune and vocal processing (IslamQA No. 391117) is the most analytically powerful analogy in this review. Scholars ruled that any digital vocal processing that omits vowels, removes phonemes, or violates Tajweed rules is impermissible. Since AI recitation tools operate on the same acoustic transformation principle, the auto-tune fatwa constitutes direct jurisprudential precedent.

2.4 Al Harere & Al Jallad (2023) — arXiv

Mispronunciation Detection of Basic Quranic Recitation Rules Using Deep Learning | arXiv:2305.06429

This paper provides the most technically granular account of AI error detection in Quranic recitation. The authors deploy LSTM neural networks on MFCC-extracted features from the QDAT dataset — a controlled

corpus of 1,500 audio samples from 150 speakers. The study targets three Tajweed rules: Separate Stretching (Mad Munfasil), Tight Noon or Meem (Ghunnah), and Concealment (Ikhfa').

The reported accuracy figures — 96%, 95%, and 96% respectively — appear impressive in isolation. However, the QDAT dataset was recorded under controlled conditions, which is categorically different from real-world recitation. Controlled-condition accuracy does not translate directly to deployment accuracy. Each misclassification is not a minor acoustic slip — it is a potential alteration of what the Words of Allah literally say.

2.5 Birgün (2026) — Computers in Human Behavior Reports

Integrating AI into Qur'an Learning: Technical Advances and Pedagogical Gaps | Computers in Human Behavior Reports

This recent paper reviews the integration of AI in Quranic education, focusing on recitation apps like Tarteel and Tajweed Mate. It highlights high accuracy in controlled settings but notes pedagogical gaps, such as the lack of spiritual guidance and over-reliance on technology without human oversight. The study emphasizes that while AI can detect errors effectively, it often fails to provide contextual explanations or adapt to individual learner needs like a human teacher.

3. Taxonomy of AI Errors in Quranic Recitation

Drawing on Papers 1, 4, and 5, the following table consolidates the principal error categories that AI recitation systems produce or fail to detect, together with their Tajweed classification and theological severity.

Error Category	Tajweed Rule Affected	Classification	Severity
Makhraj Error	Articulation points of letters	Lahn Jali	CRITICAL
Mad (Elongation) Error	Vowel stretching duration (2/4/6 counts)	Lahn Jali / Lahn Khafi	HIGH
Ghunnah Omission	Nasalisation on Noon/Meem	Lahn Khafi	MEDIUM
Ikhfa' Failure	Concealment of Noon Sakinah / Tanween	Lahn Khafi	MEDIUM
Diacritics Error	Harakat (vowel marks) on words	Lahn Jali	CRITICAL
Sequence/Skip Error	Verse order and continuity	Lahn Jali	CRITICAL
Idgham Misclassification	Merging of adjacent letters	Lahn Khafi	MEDIUM

The critical distinction embedded in this taxonomy is between Lahn Jali — manifest errors that any Arabic speaker can hear and that frequently alter lexical meaning — and Lahn Khafi — concealed errors that require specialist Tajweed knowledge to identify. Classical Islamic scholarship has universally prohibited both, but Lahn Jali is treated with greater urgency because its effects extend into semantic distortion of the Quranic text itself.

4. Quranic Evidence and the Historical Doctrine of Talaqqi

4.1 Direct Quranic Commands on Recitation Method

The Quran does not merely encourage correct recitation — it explicitly commands a specific method.

"And recite the Qur'an with measured recitation [Tarteel]."

— Quran 73:4, Surah Al-Muzzammil

The term Tarteel was defined by Sayyidna Ali ibn Abi Talib (RA) as tajweedul huruf wa ma'rifatul wuquf — the beautification of letters and the knowledge of stopping points. This definition makes clear that Tarteel is not a general aesthetic standard but a technical specification. A system that cannot consistently apply the rules of Tajweed is, by this definition, not performing Tarteel — and is therefore not fulfilling the Quranic command.

"Indeed, it is We who sent down the Quran and indeed, we will be its Guardian."

— Quran 15:9, Surah Al-Hijr

Islamic scholars have interpreted this divine guarantee of preservation as encompassing not only the textual content of the Quran but also its recitation method. The Quran is preserved in both rasm (written form) and ada' (recited form). The Talaqqi chain carries the sound of the Quran forward in time with the same fidelity that written manuscripts carry its text.

4.2 The Talaqqi-Musyafahah Doctrine and Its Implications for AI

The doctrine of Talaqqi-Musyafahah — direct, face-to-face oral transmission — is the foundational methodology of Quranic pedagogy in Islam. The Prophet received the Quran from Jibreel through oral, interactive recitation — not through writing, not through any intermediary medium. Furthermore, every Ramadan, he reviewed the entire Quran with Jibreel in a practice of annual correction and confirmation. This establishes a normative precedent of ongoing, relationship-based verification.

Following the Prophet, the companions transmitted the Quran to subsequent generations through the same methodology. The chain of transmission (Sanad) that connects contemporary reciters to the Prophet is a documented, verifiable record — every Qari who holds a formal Ijazah can trace their chain of teachers backward to the companions who received the Quran directly.

AI Systems Lack All Four Pillars of the Talaqqi Chain:

No Sanad (trained on data, not transmitted through a human lineage) • No Ijazah (no scholarly license) • No Niyyah (no conscious intention — recitation is an act of worship) • No Mutaaba'ah (static after training; cannot adapt like a living teacher)

5. Islamic Jurisprudential Analysis

5.1 The Case Against Permissibility — Majority Position

The following arguments, drawn from the five papers, primary fatwa sources, and classical Islamic principles, constitute the majority jurisprudential position that AI recitation errors are not permissible.

5.1.1 Tahreef al-Ma'na — Distortion of Quranic Meaning

Classical Tajweed scholars established that the Quran is semantically compressed to a degree unique among all texts: a single harakat (vowel mark) can reverse the grammatical role of a word, and a single elongation can transform a noun into a verb. The LSTM-based research of Al Harere & Al Jallad (2023) documents that AI systems misclassify Mad elongation at a rate that represents thousands of errors when applied across an entire Quranic recitation. Each such error is not a minor acoustic slip — it is a potential alteration of what the Words of Allah literally say.

5.1.2 The Qiyas Argument — AI Errors vs. Human Errors

Islamic jurisprudence employs Qiyas (analogy) to extend established rulings to novel cases. The established rule is that the Quran may not be learned from a person whose recitation is known to contain errors. IslamWeb Fatwa No. 510187 applies Qiyas to AI: if this prohibition applies to an erring human — who at least possesses intention, awareness, and the capacity for correction — it applies with greater force to a machine that lacks all three. This is an a fortiori argument (Qiyas bi al-awla).

5.1.3 The Auto-tune Ruling as Direct Precedent

The fatwa prohibiting auto-tune in Quranic recitation (IslamQA No. 391117) is the most directly applicable jurisprudential precedent to AI-generated or AI-processed recitation. Auto-tune and AI voice processing share the same functional characteristic: they transform the raw acoustic signal through a learned or rule-based model, producing an output that may omit phonemes, distort vowel durations, or alter the acoustic signature of specific Tajweed rules.

5.1.4 The Absence of Sanad — Foundational Disqualification

The Talaqqi doctrine establishes that authentic Quranic recitation must travel through a documented human chain. AI systems have no position in this chain and cannot be inserted into it. Their training on digital audio recordings of qualified Qaris creates a statistical approximation of correct recitation, not a faithful transmission of it. The distinction is theologically fundamental.

5.1.5 The Sincerity Exemption Does Not Apply to Machines

A frequently cited Hadith recorded in Sahih al-Bukhari states that the person who recites the Quran and falters in it, finding it difficult, receives a double reward. This hadith is sometimes invoked to suggest that AI errors might be excused. This reading is categorically incorrect. The double reward applies to the sincere human believer who struggles against phonological difficulty despite genuine effort and devotion. A machine produces errors not from struggle but from the architectural limitations of its training — it has no sincerity, no faith, no effort, and no devotion.

5.2 The Case for Conditional Permissibility — Minority Position

A minority jurisprudential position holds that AI tools are conditionally permissible, subject to the following three requirements being simultaneously satisfied.

5.2.1 Expert Scholarly Validation

IslamWeb Fatwa No. 510187 specifies that AI recitation tools may be used if they have been reviewed and endorsed by qualified Tajweed scholars as producing error-free evaluation. This is an extremely high bar:

it requires not merely that the tool has been developed with Islamic intent, but that it has been phonologically verified by scholars with formal Tajweed credentials. Currently, no publicly available AI recitation tool has undergone this level of scholarly certification.

5.2.2 Supplementary — Not Primary — Role

Adnan (2024) frames AI tools as supplements to, not replacements for, qualified human instruction. Under this condition, an AI tool may flag a potential Makhraj error — but the definitive judgment on whether an error occurred, what caused it, and how to correct it must come from a human Qari. The tool is a prompt for human correction, not a substitute for it. This mirrors the use of spell-checkers in language learning: they flag candidates, but a human speaker makes the final determination.

5.2.3 The Darura Principle — Necessity

Islamic jurisprudence contains the principle of Darura (necessity), under which what is otherwise prohibited may become temporarily permitted under conditions of genuine unavailability of the lawful alternative. For Muslims in geographic regions where no qualified Tajweed teacher is accessible — remote communities, diaspora contexts, conflict zones — a carefully validated AI tool may serve as a Darura-justified bridge to basic recitation guidance. This permission is explicitly temporary and conditional.

6. Comparative Synthesis: Evidence and Rulings

Evidence Source	Type	Position	Weight
IslamWeb Fatwa #510187	Primary Fatwa	NOT Permissible (unless expert-verified)	High
IslamQA Auto-tune #391117	Primary Fatwa (Analogy)	NOT Permissible — phonemic distortion confirmed	High
Quran 73:4 (Tarteel Command)	Quranic Nass	AI cannot consistently fulfil the Tarteel obligation	Definitive
Talaqqi-Musyafahah Doctrine	Historical Precedent	AI has no Sanad, Ijazah, or Niyyah	High
Al Harere & Al Jallad (2023)	Empirical Research	AI errors in Mad, Ghunnah, Ikhfa' documented	High
Alrumiah & Al-Shargabi (2023)	Systematic Review	Diacritics and sequence errors remain open problems	High
Adnan (2024)	Islamic-AI Review	Conditional Permit with scholar-AI collaboration	Moderate
IslamQA Fatwa #512399	Primary Fatwa	Voice permitted with consent; no recitation reward	Moderate
Darura Principle	Jurisprudential Rule	Conditional Permit under genuine inaccessibility only	Conditional
Birgün (2026)	Pedagogical Review	AI as supplement; gaps in spiritual guidance	Moderate

7. Conclusion: Majority Probability Assessment

The weight of Quranic evidence, hadith, primary fatwa rulings, classical jurisprudential principles, historical doctrine, and empirical research converges on a majority conclusion of sufficient clarity to support a probability-weighted determination.

Position	Probability	Primary Basis
AI errors NOT permissible as standard practice	~75–80%	Fatwas + Quran 73:4 + Talaqqi doctrine + auto-tune analogy + Tahreef al-Ma'na
AI tools conditionally permissible as assistive aid	~20–25%	IslamWeb conditional permit + Adnan (2024) + Darura principle
AI errors fully permissible without conditions	~0%	No credible scholarly support identified in reviewed literature

7.1 The Majority Finding

AI recitation errors are not permissible as a standard, substitutive, or primary practice in Quranic recitation. This finding rests on five independently sufficient grounds, each of which would support the conclusion even if the others were set aside:

1. The Quranic command of Tarteel, which AI systems cannot consistently fulfil
2. The Tahreef al-Ma'na principle, which prohibits any distortion of the Words of Allah
3. The Talaqqi doctrine, which requires human chain transmission that AI cannot participate in
4. The direct fatwa ruling that AI's errors disqualify it under the same principle that disqualifies erring humans
5. The auto-tune fatwa, which prohibits digital vocal processing that distorts Tajweed rules

7.2 The Qualified Minority Finding

AI tools are conditionally permissible in a narrow, carefully defined assistive role, subject to all three of the following conditions being simultaneously met: the tool has been reviewed and certified as error-free by qualified Tajweed scholars; it is used exclusively as a supplementary prompt for correction, with a human Qari retaining final authority; and it is deployed only where access to a qualified human teacher is genuinely unavailable, in accordance with the Darura principle. The moment any of these conditions' lapses, the tool falls outside the scope of conditional permissibility.

7.3 The Broader Implication

The Islamic tradition's insistence on human chain transmission in Quranic recitation is not a conservative resistance to technology — it is a principled recognition that the Quran is not merely information to be processed, but revelation to be received. The distinction matters: information can be copied, compressed, and reproduced by machines without fundamental loss of content. Revelation, in the Islamic understanding, carries an acoustic, relational, and intentional dimension that is constitutively human.

Key Conclusion:

Until AI systems can participate in the Sanad, exercise Niyyah, and be certified by scholars through the Ijazah system — none of which is theoretically or practically feasible under current understandings of AI — their errors in Quranic recitation remain, under the majority scholarly position, not permissible.

7.4 Practical Advice for Muslims

For users in regions like Rawalpindi, Pakistan, consult local scholars at institutions such as Jamia Darul Uloom or the International Islamic University Islamabad for tool validation. Always cross-verify AI feedback with a qualified Qari, and prioritize human-led learning circles (halaqat) for spiritual depth.

8. References

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"And recite the Qur'an with measured recitation." — Quran 73:4